F78AB Actuarial and Financial Mathematics

Chapter 6: Force of Interest and Continuously-paid Cashflows

6.1: The Force of Interest

- Previously we looked at:
  - Effective Rate of Interest
  - Nominal Rate of Interest
- Now we look at
  - Force of Interest

6.1: The Force of Interest

- Say you invest $A(t)$ into a fund or bank account at time t.
- No additional money is deposited and no money is withdrawn so the growth is entirely due to the interest.
- The intensity (or growth rate) is measured by the "rate of change" of the curve $A(t)$ at time t.

6.1: The Force of Interest

- But $A'(t)$ is not a good measure since it depends on $A(t)$.
- We can overcome this problem by using
- This gives us a measure of the intensity with which interest is operating at time t expressed as a rate independent of the amount of the fund i.e. **a rate per monetary unit**.

6.1: The Force of Interest

- We define this rate as the "**force of interest**"
- From this we can show that
- And for $t_2 > t_1$,

Relationship between i and δ

- If, i = effective rate of interest per time unit then 1 invested at time 0 will grow to $(1 + i)$ at time unit 1.
- If δ is the (constant) force of interest per time unit then 1 invested at time 0 will grow to e^{δ} at time unit 1.
- Hence,
- And,

Constant Force of Interest derived from the Nominal Interest Rate

- The constant force of interest, δ, can be thought of as the limit of $i^{(m)}$ as $m \to \infty$, i.e. the nominal rate of interest compounded infinitely.
- We can write this as

$$\delta = \lim_{m \to \infty} i^{(m)} = i^{(\infty)}$$

Constant Force of Interest derived from the Nominal Interest Rate

- Recall from Chapter 4 Nominal Rates of Interest
- Hence,
- Using similar arguments we can show that

Exercise 6.1 Given a constant force of interest $\delta(t) = 5\%$ p.a, for all $t \geq 0$, and assuming time is measured in years, calculate

(i) the PV at time $t = 0$ of £150 paid at time $t = 1$.

(ii) the PV at time $t = 0.5$ of £200 paid at time $t = 3.25$.

(iii) the accumulation at time $t = 4$ of £50 paid at time $t = 1$.

Exercise 6.2 Write down a formula for the annual effective interest rate i in terms of the constant force of interest δ.

More generally,

- a constant force of interest δ p.a.
- an effective rate of interest i p.a.
- a nominal rate of interest $i^{(p)}$ p.a. convertible p times per year
- a nominal rate of discount $d^{(q)}$ p.a. convertible q times per year
- an effective rate of discount d p.a.

are all equivalent if and only if the following holds:

Exercise 6.3 Write down a formula for the annual discount rate v in terms of the constant force of interest δ.

Exercise 6.4 Given that the annual effective rate of interest is 12%, calculate d, $d^{(4)}$, $d^{(12)}$, $i^{(4)}$, $i^{(12)}$ and δ.

Example 6.1.1 Find the accumulated amount at time 20 of a cash flow of £1 000 at time 0, assuming a force of interest $\delta(t)$ p.a. given by

$$\delta(t) = 0.06 + 0.0005t - 0.00005t^2.$$

Example 6.1.2 Find the accumulated amount at time 20 of a cash flow of £1 000 at time 0 and a further £1 000 at time 10, assuming a force of interest $\delta(t)$ p.a. given by

$$\delta(t) = 0.06 + 0.0005t - 0.00005t^2.$$

Exercise 6.5 *Calculate the present value at time 0 of a cashflow of 100 at time 4, and 1000 at time 10 assuming a force of interest per time unit given as follows:*

$$\delta = \begin{cases} 0.04 & 0 \le t < 5 \\ 0.14 - 0.02t & 5 \le t \end{cases}$$

6.2: Continuously-paid cashflows

- In previous sections we considered only discrete cashflows.
- Now we extend this to the case where payments are made continuously over a period of time so we have the **RATE** at which the payments are made.

6.2: Continuously-paid cashflows

- Let, $M(t)$ = TOTAL payment made between tine 0 and time $t > 0$.
- The rate of payment is
- For $0 < t_1 < t_2$, the TOTAL payment between time t_1 and time t_2 is

Example 6.2.1:

Notation for Present/Discounted Values and Accumulated Values

- For times s and t, where $0 < s < t$, we denote
- $A(s,t)$ as the accumulated value at time t of 1 paid at time s.
- $V(s,t)$ as the present/discounted value at time s of 1 paid at time t.

Example 6.2.2:

Present Values

- Consider a cashflow paid continuously at the rate of $\rho(t) = M'(t)$ per annum for T years starting at time 0.
- Partition the time interval $[0, T]$ into tiny disjoint segments, each of length dt. Consider one of these tiny segments $[t + dt)$.
- If dt is very small, then the total payment between time t and time $t + dt$ is

Present Values

- We treat the amount paid between t and $t + dt$ as paid exactly at time t, i.e. we treat the continuous payment made over $[t + dt)$ as a discrete cashflow $\rho(t)dt$ paid at time t.
- The PV of this discrete cashflow is $V(0, t)\rho(t)dt$.
- Repeat for all the tiny segments intervals, then sum them to find the total PV at time 0.

Present Values

- Letting $dt \to 0$, we find in the limit that the present value at time 0 of the entire cashflow between time 0 and time T is

Accumulated Values

- For the accumulation at time T we follow a similar argument as for the present value.
- Consider a cashflow paid continuously at the rate $\rho(t) = M'(t)$ per annum for $(T - S)$ years starting at time S.
- The accumulated value at time T of the entire cashflow between time S and time T is

Different types of cashflows The accumulation at time $T > 0$ of

1. one discrete payment $C(t_1)$ made at time t_1 is

$$A(T) = C(t_1) \exp\left(\int_{t_1}^{T} \delta(s)ds\right).$$

2. n discrete payments $C(t_k)$ made at times t_k, for $k = 1, \ldots, n$ is

$$A(T) = \sum_{k=1}^{n} C(t_k) \exp\left(\int_{t_k}^{T} \delta(s)ds\right).$$

3. a continuous payment paid at rate $\rho(t)$ for all $t \in [S, T]$, for some $S < T$, is

$$A(T) = \int_{S}^{T} \rho(t) \exp\left(\int_{t}^{T} \delta(u)du\right) dt.$$

International Actuarial Notation for Continuously Payable Annuities

- Consider an annuity with a constant rate of £1 per year, paid continuously during the year, for n years.
- We assume that the force of interest is constant, $\delta(t) = \delta, \forall t \geq 0$.
- We might interpret this as the limit of an annuity paid m times each year, i.e.

International Actuarial Notation for Continuously Payable Annuities

- Another way to get $\bar{a}_{\overline{n}|}$ is from the present/discount equation above

International Actuarial Notation for Continuously Payable Annuities

- The value at time n of this continuously payable annuity is

International Actuarial Notation for Continuously Payable Annuities

- Another way to get $\bar{s}_{\overline{n}|}$ is from the accumulation equation above

Present Values

- Now we look at some cases for present values based on the formula:

- **Case 1:** $\rho(t) = C$ (constant).
- **Case 2:** $\delta(t) = \delta$ (constant).
- **Case 3:** $\rho(t) = C$ AND $\delta(t) = \delta$.

Examples and Exercises

- Example 6.2.3
- Exercise 6.6
- Example 6.2.4
- Exercise 6.7
- Example 6.2.5
- Exercise 6.8
- Exercise 6.9
- Exercise 6.10
- Example 6.2.6
- Example 6.2.7

Increasing Annuity

- Rate of payment k units per annum in year k.
- The PV at the start of the cashflow is
- The AV at time n is

Increasing Annuity

- Rate of payment t units per annum at time t.
- The PV at the start of the cashflow is
- The AV at time n is

Examples and Exercises

- Example 6.2.8:
- Exercise 6.11:
- Exercise 6.12:

Next Chapter

Chapter 7: Loans and Loan Schedules